

QuizGen AI

Evaluation Report

Intelligent Reading Comprehension & Quiz Generation System
National University of Computer and Emerging Sciences · AI Lab · Spring 2026

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Dataset Overview

RACE (ReAding Comprehension from Examinations) — Lai et al., EMNLP 2017. English multiple-choice questions from Chinese middle and high school exams.

Split	MCQ Rows	Binary Samples	Feature Dims
Train	70,292	281,168	6,008
Val	8,787	35,148	6,008
Test	8,787	35,148	6,008

Feature pipeline: TF-IDF (unigrams+bigrams, 6000 vocab, sublinear_tf) + 8 dense features (lengths, overlap ratios, cosine similarities) → hstack → StandardScaler.

Model A — Answer Verifier

Binary classification formulation: each MCQ → 4 samples (article, question, option, label). At inference, all 4 options are scored and the highest-probability option is selected. Random baseline = 25.0%.

Supervised Model Comparison

Model	Val Acc	Val F1	Test Acc	Test F1	vs Baseline
Logistic Regression	37.1%	37.1%	37.0%	37.0%	+12.0%
Linear SVM	37.3%	37.2%	37.1%	37.1%	+12.1%
Complement NB	32.6%	32.5%	32.2%	32.2%	+7.2%
Random Forest	35.3%	35.3%	35.4%	35.3%	+10.3%
XGBoost	36.1%	36.0%	36.4%	36.4%	+11.4%
Soft-Vote Ensemble	37.3%	37.3%	37.4%	37.4%	+12.4%
Random Baseline	25.0%	25.0%	25.0%	25.0%	—

★ Best model: Soft-Vote Ensemble (highlighted in green)

Confusion Matrix — Soft-Vote Ensemble (Test Set)

	Pred A	Pred B	Pred C	Pred D
True A	737	393	376	409
True B	489	822	451	510
True C	510	455	892	532
True D	489	436	449	837

Classification Report (Test Set)

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precision recall f1-score support
A 0.33 0.38 0.36 1915
B 0.39 0.36 0.38 2272
C 0.41 0.37 0.39 2389
D 0.37 0.38 0.37 2211
accuracy 0.37 8787
macro avg 0.37 0.37 0.37 8787
weighted avg 0.38 0.37 0.37 8787
```

Model A — Unsupervised & Semi-Supervised

Evaluated on OHE features (5000-dim binary bag-of-words). K-Means groups question-answer pairs into 4 clusters. Label Spreading uses 15% labeled samples to propagate labels.

Method	Metric	Value	Notes
K-Means (k=4)	Cluster Purity	0.2720	Majority-class fraction per cluster
K-Means (k=4)	Silhouette Score	0.0000	Cluster separation quality
Label Spreading	Val Accuracy	0.2775	Evaluated on 2,000 samples
Label Spreading	Val Macro-F1	0.1095	vs supervised best: 0.3733

Model B — Distractor Ranker

XGBoost binary classifier trained on phrase-based candidates with hard negative sampling. Features: TF-IDF cosine similarities (candidate↔answer, candidate↔question, candidate↔article), frequency, length, word overlap, multi-word flag.

Split	Accuracy	Precision	Recall	F1	NDCG@3
Val	95.5%	98.9%	92.0%	95.3%	0.9701
Test	95.7%	98.2%	93.1%	95.6%	0.9761

Model B — Hint Scorer

Logistic Regression trained on sentence-level features. Labels assigned by similarity (top-2 sentences per article by TF-IDF cosine sim to question). Precision@3: fraction of top-3 ranked sentences that are truly relevant.

Split	Accuracy	Precision	Recall	F1	Precision@3
Val	81.2%	46.5%	73.0%	56.8%	100.0%
Test	81.1%	45.9%	69.0%	55.1%	100.0%

Executive Summary

Component	Key Result	Status
Model A (best)	Soft-Vote Ensemble — Test Acc 37.4%	✓ Above random (25%)
Model A (ensemble)	Soft-Vote LR+RF — 37.4%	✓ Best overall
K-Means clustering	Purity 0.2720	✓ Evaluated
Label Spreading	Val F1 0.1095	✓ Evaluated
Distractor Ranker	Test Acc 95.7% NDCG@3 0.9761	✓ High quality
Hint Scorer	Test Acc 81.1% P@3 100.0%	✓ Evaluated